

```
import java.util.Scanner;

class Book {
    private int bookId;
    private String title;
    private String author;
    private double price;
    private boolean issued;

    Book(int id, String t, String a, double p) {
        bookId = id;
        title = t;
        author = a;
        price = p;
        issued = false;
    }

    void display() {
        System.out.println("Book ID : " + bookId);
        System.out.println("Title : " + title);
        System.out.println("Author : " + author);
        System.out.println("Price : " + price);
        System.out.println("Status : " + (issued ? "Issued" : "Available"));
    }

    void issueBook() {
        if (!issued) {
            issued = true;
            System.out.println("Book Issued Successfully.");
        } else {
            System.out.println("Book Already Issued.");
        }
    }
}
```

```

    }
}

void returnBook() {
    if (issued) {
        issued = false;

        System.out.println("Book Returned Successfully.");
    } else {
        System.out.println("Book is Already Available.");
    }
}

int getBookId() {
    return bookId;
}
}

```

```

public class LibraryManagement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        Book[] books = new Book[5];

        int count = 0;

        int choice;

        do {
            System.out.println("\n----- Library Menu -----");

            System.out.println("1. Add Book");

            System.out.println("2. Display Books");

            System.out.println("3. Issue Book");

            System.out.println("4. Return Book");

            System.out.println("5. Search Book");

```

```
System.out.println("6. Exit");

System.out.print("Enter Choice: ");

choice = sc.nextInt();

switch (choice) {

    case 1:

        System.out.print("Enter Book ID: ");

        int id = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Title: ");

        String title = sc.nextLine();

        System.out.print("Enter Author: ");

        String author = sc.nextLine();

        System.out.print("Enter Price: ");

        double price = sc.nextDouble();

        books[count] = new Book(id, title, author, price);

        count++;

        System.out.println("Book Added Successfully.");

        break;

    case 2:

        for (int i = 0; i < count; i++) {

            books[i].display();

            System.out.println();

        }

        break;

    case 3:

        System.out.print("Enter Book ID: ");

        id = sc.nextInt();

        for (int i = 0; i < count; i++) {
```

```
        if (books[i].getBookId() == id) {  
            books[i].issueBook();  
        }  
    }  
    break;
```

case 4:

```
    System.out.print("Enter Book ID: ");  
    id = sc.nextInt();  
    for (int i = 0; i < count; i++) {  
        if (books[i].getBookId() == id) {  
            books[i].returnBook();  
        }  
    }  
    break;
```

case 5:

```
    System.out.print("Enter Book ID: ");  
    id = sc.nextInt();  
    boolean found = false;  
    for (int i = 0; i < count; i++) {  
        if (books[i].getBookId() == id) {  
            books[i].display();  
            found = true;  
        }  
    }  
    if (!found) {  
        System.out.println("Book Not Found.");  
    }  
    break;
```

case 6:

```
System.out.println("Thank You!");
```

```
break;
```

default:

```
System.out.println("Invalid Choice.");
```

```
}
```

```
} while (choice != 6);
```

```
sc.close();
```

```
}
```

```
}
```

The screenshot displays the Programiz Online Java Compiler interface. The code editor on the left contains a Java program with line numbers 157 to 184. The program includes a search function, a menu, and a while loop that continues until the user enters choice 6. The output window on the right shows the execution results, including the menu display, user input for book details, and the successful addition of a book.

```
157 boolean foundSearch = false;
158
159 for (int i = 0; i < count; i++) {
160     if (books[i].getBookId() == id) {
161         books[i].display();
162         foundSearch = true;
163         break;
164     }
165 }
166
167 if (!foundSearch) {
168     System.out.println("Book Not Found.");
169 }
170 break;
171
172 case 6:
173     System.out.println("Thank You!");
174     break;
175
176 default:
177     System.out.println("Invalid Choice.");
178 }
179
180 } while (choice != 6);
181
182 sc.close();
183 }
184 }
```

Output:

```
4. Return Book
5. Search Book
6. Exit
Enter Choice:
1
Enter Book ID: 101
Enter Title: java programing
Enter Author: James
Enter Price: 500
Book Added Successfully.

===== Library Menu =====
1. Add Book
2. Display Books
3. Issue Book
4. Return Book
5. Search Book
6. Exit
Enter Choice: 2
Book ID : 101
Title : java programming
Author : James
Price : 500.0
Status : Available
-----

===== Library Menu =====
1. Add Book
2. Display Books
```

```
2 class Employee {
    String name;
    Employee(String name) {
        this.name = name;
    }
    void calculateSalary() {
        System.out.println("Salary");
    }
}

class PermanentEmployee extends Employee {
    PermanentEmployee(String name) {
        super(name);
    }
    void calculateSalary() {
        System.out.println(name + " Salary = 50000");
    }
}

class ContractEmployee extends Employee {
    ContractEmployee(String name) {
        super(name);
    }
    void calculateSalary() {
        System.out.println(name + " Salary = 25000");
    }
}

public class Main {
    public static void main(String[] args) {
        Employee e1 = new PermanentEmployee("Ravi");
    }
}
```

```

        Employee e2 = new ContractEmployee("Rahul");

        e1.calculateSalary();

        e2.calculateSalary();

    }

}

```

```

Main.java
17     }
18
19     void calculateSalary() {
20         System.out.println(name + " Salary = 50000");
21     }
22 }
23
24 class ContractEmployee extends Employee {
25
26     ContractEmployee(String name) {
27         super(name);
28     }
29
30     void calculateSalary() {
31         System.out.println(name + " Salary = 25000");
32     }
33 }
34
35 public class Main {
36     public static void main(String[] args) {
37
38         Employee e1 = new PermanentEmployee("Ravi");
39         Employee e2 = new ContractEmployee("Rahul");
40
41         e1.calculateSalary();
42         e2.calculateSalary();
43     }
}

Output
* Ravi Salary = 50000
Rahul Salary = 25000

=== Code Execution Successful ===

```

3.

```
import java.util.Scanner;
```

```

class Vehicle {

    void calculateRent(int days) {

        System.out.println("Vehicle Rent");

    }

}

```

```

class Car extends Vehicle {

    void calculateRent(int days) {

        System.out.println("Car Rent = Rs." + (days * 1000));

    }

}

```

```
class Bike extends Vehicle {  
    void calculateRent(int days) {  
        System.out.println("Bike Rent = Rs." + (days * 300));  
    }  
}
```

```
class Bus extends Vehicle {  
    void calculateRent(int days) {  
        System.out.println("Bus Rent = Rs." + (days * 2000));  
    }  
}
```

```
public class Main {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        Vehicle[] vehicles = new Vehicle[3];  
  
        vehicles[0] = new Car();  
        vehicles[1] = new Bike();  
        vehicles[2] = new Bus();  
  
        System.out.println("Vehicle Rental System");  
        System.out.println("1. Car");  
        System.out.println("2. Bike");  
        System.out.println("3. Bus");  
    }  
}
```



```
System.out.print("Enter your choice: ");
```

```
int choice = sc.nextInt();
```

```
System.out.print("Enter number of days: ");
```

```
int days = sc.nextInt();
```

```
if (choice >= 1 && choice <= 3) {
```

```
    vehicles[choice - 1].calculateRent(days);
```

```
} else {
```

```
    System.out.println("Invalid Choice");
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```

```
Main.java  Run  Output  Clear
30 Scanner sc = new Scanner(System.in);
31
32 Vehicle[] vehicles = new Vehicle[3];
33
34 vehicles[0] = new Car();
35 vehicles[1] = new Bike();
36 vehicles[2] = new Bus();
37
38 System.out.println("Vehicle Rental System");
39 System.out.println("1. Car");
40 System.out.println("2. Bike");
41 System.out.println("3. Bus");
42
43 System.out.print("Enter your choice: ");
44 int choice = sc.nextInt();
45
46 System.out.print("Enter number of days: ");
47 int days = sc.nextInt();
48
49 if (choice >= 1 && choice <= 3) {
50     vehicles[choice - 1].calculateRent(days);
51 } else {
52     System.out.println("Invalid Choice");
53 }
54
55 sc.close();
56 }
57 }
```

```
Vehicle Rental System
1. Car
2. Bike
3. Bus
Enter your choice: 1
Enter number of days: 5
Car Rent = Rs.5000

=== Code Execution Successful ===
```

4. import java.util.Scanner;

class BankAccount {

// Private data members (Data Hiding)

private int accountNumber;

private String holderName;

private double balance;

// Constructor

BankAccount(int accNo, String name, double bal) {

accountNumber = accNo;

holderName = name;

balance = bal;

}

// Deposit method

void deposit(double amount) {

if (amount > 0) {

balance = balance + amount;

System.out.println("Amount Deposited Successfully.");

} else {

System.out.println("Invalid Deposit Amount.");

}

}

// Withdraw method

void withdraw(double amount) {

```
    if (amount <= balance) {  
        balance = balance - amount;  
        System.out.println("Amount Withdrawn Successfully.");  
    } else {  
        System.out.println("Insufficient Balance.");  
    }  
}
```

```
// Display Account Details
```

```
void display() {  
    System.out.println("\nAccount Number : " + accountNumber);  
    System.out.println("Holder Name   : " + holderName);  
    System.out.println("Balance      : Rs." + balance);  
}  
}
```

```
public class Main {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter Account Number: ");  
        int accNo = sc.nextInt();  
        sc.nextLine();  
  
        System.out.print("Enter Holder Name: ");  
        String name = sc.nextLine();
```

```
System.out.print("Enter Initial Balance: ");
```

```
double balance = sc.nextDouble();
```

```
BankAccount account = new BankAccount(accNo, name, balance);
```

```
System.out.println("\n1. Deposit");
```

```
System.out.println("2. Withdraw");
```

```
System.out.println("3. Display Balance");
```

```
System.out.print("Enter Choice: ");
```

```
int choice = sc.nextInt();
```

```
switch (choice) {
```

```
    case 1:
```

```
        System.out.print("Enter Deposit Amount: ");
```

```
        double deposit = sc.nextDouble();
```

```
        account.deposit(deposit);
```

```
        account.display();
```

```
        break;
```

```
    case 2:
```

```
        System.out.print("Enter Withdraw Amount: ");
```

```
        double withdraw = sc.nextDouble();
```

```
        account.withdraw(withdraw);
```

```
        account.display();
```

```
        break;
```

```

        case 3:

            account.display();

            break;

        default:

            System.out.println("Invalid Choice");

    }

    sc.close();

}

}

```

The screenshot shows a Java IDE with a file named `Main.java`. The code implements a banking system with a `switch` statement for user choices. The output window on the right shows the execution flow: entering account details (101, ravi, 5000), selecting '1' for deposit, entering 1000, and displaying the updated balance of Rs. 6000.0. The execution is successful.

```

Main.java
64 System.out.println("3. Display Balance");
65 System.out.print("Enter Choice: ");
66
67 int choice = sc.nextInt();
68
69 switch (choice) {
70
71     case 1:
72         System.out.print("Enter Deposit Amount: ");
73         double deposit = sc.nextDouble();
74         account.deposit(deposit);
75         account.display();
76         break;
77
78     case 2:
79         System.out.print("Enter Withdraw Amount: ");
80         double withdraw = sc.nextDouble();
81         account.withdraw(withdraw);
82         account.display();
83         break;
84
85     case 3:
86         account.display();
87         break;
88
89     default:
90

```

Output

```

* Enter Account Number: 101
Enter Holder Name: ravi
Enter Initial Balance: 5000

1. Deposit
2. Withdraw
3. Display Balance
Enter Choice: 1
Enter Deposit Amount: 1000
Amount Deposited Successfully.

Account Number : 101
Holder Name    : ravi
Balance       : Rs.6000.0

=== Code Execution Successful ===

```

```
5. import java.util.Scanner;
```

```
class Admission {
```

```
    // Undergraduate Fee
```

```
    void calculateFee() {
```

```
        System.out.println("Course : Undergraduate");
```

```
        System.out.println("Fee = Rs.50000");
```

```
    }
```

```
    // Postgraduate Fee
```

```
    void calculateFee(int pg) {
```

```
        System.out.println("Course : Postgraduate");
```

```
        System.out.println("Fee = Rs.70000");
```

```
    }
```

```
    // Scholarship Student Fee
```

```
    void calculateFee(double scholarship) {
```

```
        System.out.println("Course : Scholarship");
```

```
        System.out.println("Fee = Rs.20000");
```

```
    }
```

```
}
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
Admission a = new Admission();

System.out.println("University Admission System");
System.out.println("1. Undergraduate");
System.out.println("2. Postgraduate");
System.out.println("3. Scholarship Student");

System.out.print("Enter your choice: ");
int choice = sc.nextInt();

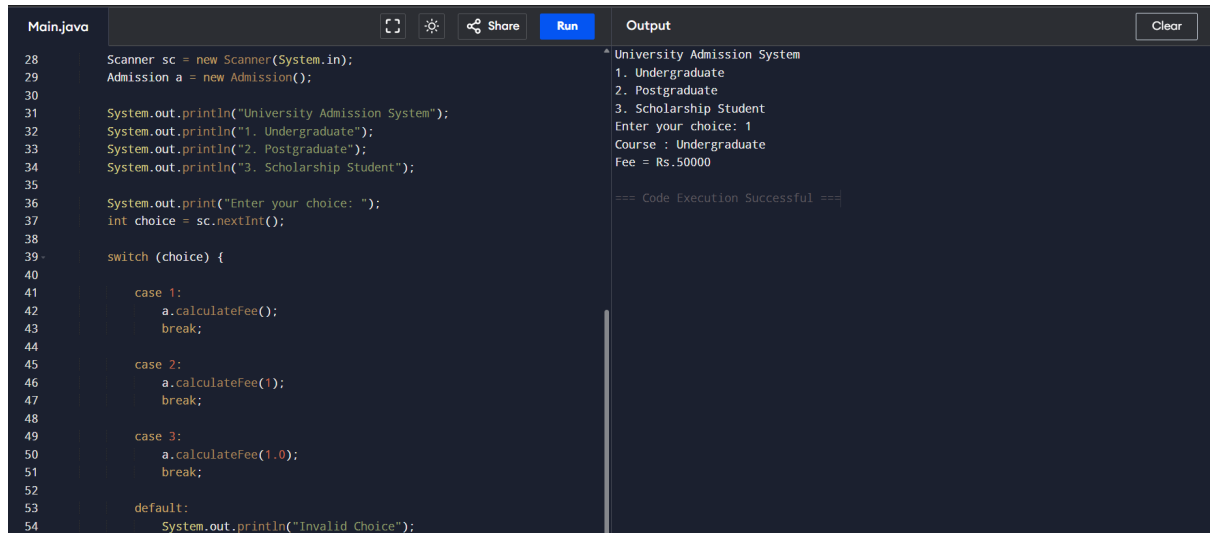
switch (choice) {

    case 1:
        a.calculateFee();
        break;

    case 2:
        a.calculateFee(1);
        break;
    case 3:
        a.calculateFee(1.0);
        break;
    default:
        System.out.println("Invalid Choice");
}

sc.close();
}
```

```
}
```



```
Main.java
28 Scanner sc = new Scanner(System.in);
29 Admission a = new Admission();
30
31 System.out.println("University Admission System");
32 System.out.println("1. Undergraduate");
33 System.out.println("2. Postgraduate");
34 System.out.println("3. Scholarship Student");
35
36 System.out.print("Enter your choice: ");
37 int choice = sc.nextInt();
38
39 switch (choice) {
40
41     case 1:
42         a.calculateFee();
43         break;
44
45     case 2:
46         a.calculateFee(1);
47         break;
48
49     case 3:
50         a.calculateFee(1.0);
51         break;
52
53     default:
54         System.out.println("Invalid Choice");
}

Output
^ University Admission System
1. Undergraduate
2. Postgraduate
3. Scholarship Student
Enter your choice: 1
Course : Undergraduate
Fee = Rs.50000

=== Code Execution Successful ===
```

```
6. import java.util.Scanner;
```

```
abstract class Shape {
```

```
    abstract void area();
```

```
}
```

```
class Circle extends Shape {
```

```
    double radius;
```

```
    Circle(double r) {
```

```
        radius = r;
```

```
    }
```

```
    void area() {
```

```
        double a = 3.14 * radius * radius;
```

```
        System.out.println("Area of Circle = " + a);
```

```
    }
```

```
}
```

```
class Rectangle extends Shape {
```

```
    double length, width;
```

```
    Rectangle(double l, double w) {
```

```
        length = l;
```



```

        width = w;
    }

    void area() {
        double a = length * width;
        System.out.println("Area of Rectangle = " + a);
    }
}

class Triangle extends Shape {
    double base, height;

    Triangle(double b, double h) {
        base = b;
        height = h;
    }

    void area() {
        double a = 0.5 * base * height;
        System.out.println("Area of Triangle = " + a);
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Shape[] shapes = new Shape[3];
        System.out.print("Enter Radius of Circle: ");
        double r = sc.nextDouble();
        System.out.print("Enter Length of Rectangle: ");
        double l = sc.nextDouble();
    }
}

```

```

System.out.print("Enter Width of Rectangle: ");

double w = sc.nextDouble();

System.out.print("Enter Base of Triangle: ");

double b = sc.nextDouble();

System.out.print("Enter Height of Triangle: ");

double h = sc.nextDouble();

shapes[0] = new Circle(r);

shapes[1] = new Rectangle(l, w);

shapes[2] = new Triangle(b, h);

System.out.println("\nAreas of Shapes:");

for (Shape s : shapes) {

    s.area();

}

sc.close();

}

```

The screenshot shows a Java IDE with a file named 'Main.java'. The code in the editor is as follows:

```

57 Scanner sc = new Scanner(System.in);
58
59 Shape[] shapes = new Shape[3];
60
61 System.out.print("Enter Radius of Circle: ");
62 double r = sc.nextDouble();
63
64 System.out.print("Enter Length of Rectangle: ");
65 double l = sc.nextDouble();
66
67 System.out.print("Enter Width of Rectangle: ");
68 double w = sc.nextDouble();
69
70 System.out.print("Enter Base of Triangle: ");
71 double b = sc.nextDouble();
72
73 System.out.print("Enter Height of Triangle: ");
74 double h = sc.nextDouble();
75
76 shapes[0] = new Circle(r);
77 shapes[1] = new Rectangle(l, w);
78 shapes[2] = new Triangle(b, h);

```

The IDE has a 'Run' button and a 'Share' button. The 'Output' window on the right shows the following text:

```

Enter Radius of Circle: 5
Enter Length of Rectangle: 10
Enter Width of Rectangle: 4
Enter Base of Triangle: 8
Enter Height of Triangle: 6

Areas of Shapes:
Area of Circle = 78.5
Area of Rectangle = 40.0
Area of Triangle = 24.0

=== Code Execution Successful ===

```

```
7. import java.util.Scanner;

interface MedicalRecord {

    void addRecord();

    void displayRecord();

}

class Patient implements MedicalRecord {

    int patientId;

    String patientName;

    Patient(int id, String name) {

        patientId = id;

        patientName = name;

    }

    public void addRecord() {

        System.out.println("Patient Record Added Successfully.");

    }


    public void displayRecord() {

        System.out.println("\nPatient Details");

        System.out.println("Patient ID : " + patientId);

        System.out.println("Patient Name : " + patientName);

    }

}

class Doctor implements MedicalRecord {

    int doctorId;

    String doctorName;

    Doctor(int id, String name) {

        doctorId = id;

        doctorName = name;

    }

}
```

```

    }

    public void addRecord() {
        System.out.println("Doctor Record Added Successfully.");
    }

    public void displayRecord() {
        System.out.println("\nDoctor Details");
        System.out.println("Doctor ID : " + doctorId);
        System.out.println("Doctor Name : " + doctorName);
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Hospital Management System");
        System.out.println("1. Patient");
        System.out.println("2. Doctor");
        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();
        sc.nextLine();
        if (choice == 1) {
            System.out.print("Enter Patient ID: ");
            int id = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Patient Name: ");
            String name = sc.nextLine();
            Patient p = new Patient(id, name);
            p.addRecord();
        }
    }
}

```

```

        p.displayRecord();
    } else if (choice == 2) {
        System.out.print("Enter Doctor ID: ");

        int id = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Doctor Name: ");

        String name = sc.nextLine();

        Doctor d = new Doctor(id, name);

        d.addRecord();

        d.displayRecord();
    } else {

        System.out.println("Invalid Choice");
    }

    sc.close();
}
}

```

The screenshot displays a Java IDE with a file named 'Main.java'. The code is a Java program for a Hospital Management System. It includes a menu with options for Patient and Doctor, input for choice, patient ID, and name, and logic to add and display records. The output window shows the program's execution, including the menu display, choice input (1), patient ID input (101), patient name input (jyothi), and the successful addition of the patient record. The output also shows the patient details: Patient ID : 101 and Patient Name : jyothi. The execution ends with '=== Code Execution Successful ==='.

```

Main.java
69
70     System.out.print("Enter Patient ID: ");
71     int id = sc.nextInt();
72     sc.nextLine();
73
74     System.out.print("Enter Patient Name: ");
75     String name = sc.nextLine();
76
77     Patient p = new Patient(id, name);
78     p.addRecord();
79     p.displayRecord();
80
81 } else if (choice == 2) {
82
83     System.out.print("Enter Doctor ID: ");
84     int id = sc.nextInt();
85     sc.nextLine();
86
87     System.out.print("Enter Doctor Name: ");
88     String name = sc.nextLine();
89
90     Doctor d = new Doctor(id, name);
91     d.addRecord();
92     d.displayRecord();
93
94 } else {

```

Output

```

Hospital Management System
1. Patient
2. Doctor
Enter Choice: 1
Enter Patient ID: 101
Enter Patient Name: jyothi
Patient Record Added Successfully.

Patient Details
Patient ID   : 101
Patient Name : jyothi

=== Code Execution Successful ===

```

```
8. import java.util.Scanner;

class Product {

    String productName;

    int price;

    Product(String name, int price) {

        this.productName = name;

        this.price = price;

    }

}

class Order {

    void placeOrder(Product p, int quantity) {

        int total = p.price * quantity;

        System.out.println("\n----- Shopping Bill -----");

        System.out.println("Product Name : " + p.productName);

        System.out.println("Price      : Rs." + p.price);

        System.out.println("Quantity   : " + quantity);

        System.out.println("Total Amount : Rs." + total);

    }

}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Online Shopping Order System");
```

```
System.out.print("Enter Product Name: ");
```

```
String name = sc.nextLine();
```

```
System.out.print("Enter Product Price: ");
```

```
int price = sc.nextInt();
```

```
System.out.print("Enter Quantity: ");
```

```
int quantity = sc.nextInt();
```

```
Product p = new Product(name, price);
```

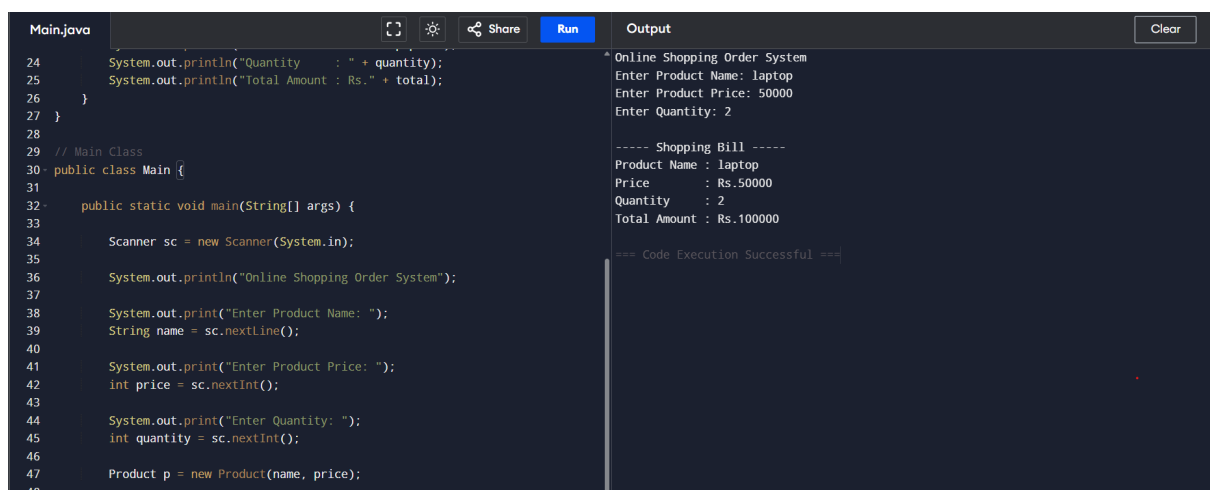
```
Order o = new Order();
```

```
o.placeOrder(p, quantity);
```

```
sc.close();
```

```
}
```

```
}
```



The screenshot shows a Java IDE with a file named 'Main.java'. The code in the editor includes a main method that prompts the user for product name, price, and quantity, then calculates the total amount. The output window on the right shows the execution results, including the product name 'laptop', price '50000', quantity '2', and a total amount of 'Rs.100000'. The output also displays a shopping bill and a success message.

```
Main.java
24 System.out.println("Quantity : " + quantity);
25 System.out.println("Total Amount : Rs." + total);
26 }
27 }
28
29 // Main Class
30 public class Main {
31
32     public static void main(String[] args) {
33
34         Scanner sc = new Scanner(System.in);
35
36         System.out.println("Online Shopping Order System");
37
38         System.out.print("Enter Product Name: ");
39         String name = sc.nextLine();
40
41         System.out.print("Enter Product Price: ");
42         int price = sc.nextInt();
43
44         System.out.print("Enter Quantity: ");
45         int quantity = sc.nextInt();
46
47         Product p = new Product(name, price);
48     }
49 }
```

```
Output
* Online Shopping Order System
Enter Product Name: laptop
Enter Product Price: 50000
Enter Quantity: 2

----- Shopping Bill -----
Product Name : laptop
Price : Rs.50000
Quantity : 2
Total Amount : Rs.100000

=== Code Execution Successful ===
```

```
9. import java.util.Scanner;
```

```
class Student {  
    int rollNo;  
    String name;  
    int m1, m2, m3;  
    int total;  
    double average;  
    String grade;  
    Student(int rollNo, String name, int m1, int m2, int m3) {  
        this.rollNo = rollNo;  
        this.name = name;  
        this.m1 = m1;  
        this.m2 = m2;  
        this.m3 = m3;  
    }  
    void calculateResult() {  
        total = m1 + m2 + m3;  
        average = total / 3.0;  
  
        if (average >= 90)  
            grade = "A";  
        else if (average >= 75)  
            grade = "B";  
        else if (average >= 60)  
            grade = "C";  
        else  
            grade = "Fail";  
    }  
}
```



```

    }

    void display() {
        System.out.println("\nRoll No : " + rollNo);
        System.out.println("Name   : " + name);
        System.out.println("Total   : " + total);
        System.out.println("Average : " + average);
        System.out.println("Grade   : " + grade);
    }
}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Number of Students: ");
        int n = sc.nextInt();
        Student[] students = new Student[n];
        for (int i = 0; i < n; i++) {
            System.out.println("\nEnter Details of Student " + (i + 1));
            System.out.print("Roll No: ");
            int roll = sc.nextInt();
            sc.nextLine();
            System.out.print("Name: ");
            String name = sc.nextLine();
            System.out.print("Marks in Subject 1: ");
            int m1 = sc.nextInt();
            System.out.print("Marks in Subject 2: ");
            int m2 = sc.nextInt();
            System.out.print("Marks in Subject 3: ");

```

```

        int m3 = sc.nextInt();

        students[i] = new Student(roll, name, m1, m2, m3);
        students[i].calculateResult();
    }

    System.out.println("\n----- Student Results -----");

    int maxTotal = students[0].total;
    String topper = students[0].name;

    for (int i = 0; i < n; i++) {
        students[i].display();

        if (students[i].total > maxTotal) {
            maxTotal = students[i].total;
            topper = students[i].name;
        }
    }

    System.out.println("\nClass Topper : " + topper);
    System.out.println("Top Score   : " + maxTotal);

    sc.close();
}
}

```

The screenshot shows a Java IDE with a file named 'Main.java'. The code in the editor matches the code block above. The IDE has a 'Run' button and a 'Share' icon. The 'Output' window on the right shows the following text:

```

Name: kiran
Marks in Subject 1: 90
Marks in Subject 2: 50
Marks in Subject 3: 80

Enter Details of Student 2
Roll No: 102
Name: 60
Marks in Subject 1: 90
Marks in Subject 2: 90
Marks in Subject 3: 100

----- Student Results -----

Roll No : 101
Name   : kiran
Total  : 220
Average : 73.33333333333333
Grade  : C

Roll No : 102
Name   : 60
Total  : 280
Average : 93.33333333333333
Grade  : A

Class Topper : 60
Top Score   : 280

```

```
10. import java.util.Scanner;

class Device {
    void turnOn() {
        System.out.println("Device is ON");
    }
    void turnOff() {
        System.out.println("Device is OFF");
    }
    void displayPower() {
        System.out.println("Power Consumption");
    }
}

class Light extends Device {
    void displayPower() {
        System.out.println("Light Power Consumption = 20 Watts");
    }
}

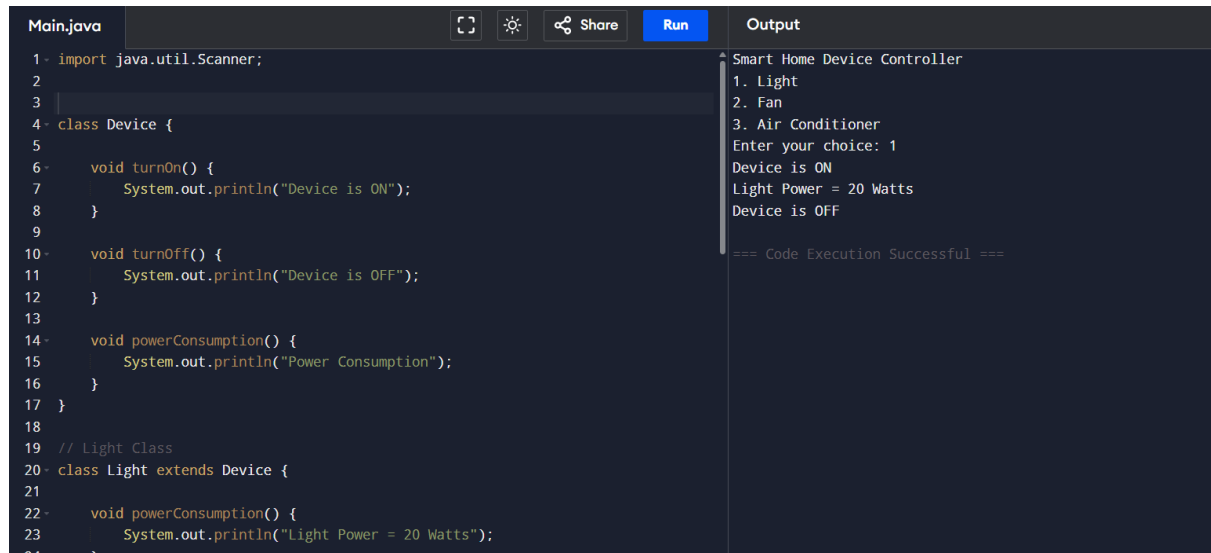
class Fan extends Device {

    void displayPower() {
        System.out.println("Fan Power Consumption = 75 Watts");
    }
}

class AirConditioner extends Device {
    void displayPower() {
        System.out.println("Air Conditioner Power Consumption = 1500 Watts");
    }
}
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        Device device = null;  
        System.out.println("===== Smart Home Device Controller =====");  
        System.out.println("1. Light");  
        System.out.println("2. Fan");  
        System.out.println("3. Air Conditioner");  
        System.out.print("Enter your choice: ");  
        int choice = sc.nextInt();  
        switch (choice) {  
            case 1:  
                device = new Light();  
                break;  
            case 2:  
                device = new Fan();  
                break;  
            case 3:  
                device = new AirConditioner();  
                break;  
            default:  
                System.out.println("Invalid Choice");  
                return;  
        }  
        device.turnOn();  
        device.displayPower();  
        device.turnOff();  
    }  
}
```

```
        sc.close();
    }
}
```



The screenshot shows a Java IDE with a file named 'Main.java'. The code defines a base class 'Device' and a subclass 'Light'. The 'Device' class has methods 'turnOn()', 'turnOff()', and 'powerConsumption()'. The 'Light' class inherits from 'Device' and overrides 'powerConsumption()' to print 'Light Power = 20 Watts'. The IDE interface includes a 'Run' button and an 'Output' pane. The output shows the program's execution, including a menu of options (1. Light, 2. Fan, 3. Air Conditioner), user input '1', and the resulting output messages: 'Smart Home Device Controller', 'Device is ON', 'Light Power = 20 Watts', and 'Device is OFF'. The execution concludes with '=== Code Execution Successful ==='.

```
Main.java
1- import java.util.Scanner;
2-
3-
4- class Device {
5-
6-     void turnOn() {
7-         System.out.println("Device is ON");
8-     }
9-
10-    void turnOff() {
11-        System.out.println("Device is OFF");
12-    }
13-
14-    void powerConsumption() {
15-        System.out.println("Power Consumption");
16-    }
17- }
18-
19- // Light Class
20- class Light extends Device {
21-
22-     void powerConsumption() {
23-         System.out.println("Light Power = 20 Watts");
24-     }
25- }
```

Output

```
Smart Home Device Controller
1. Light
2. Fan
3. Air Conditioner
Enter your choice: 1
Device is ON
Light Power = 20 Watts
Device is OFF

=== Code Execution Successful ===
```